

Catching Anomalous Providers in Medicare Claims: An Unsupervised Screen for Payment Integrity

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Abstract

Medicare loses billions every year to improper payments, and a large share of the review work is still finding which providers are worth a second look. I treat that as an anomaly-detection problem: instead of training on confirmed fraud, score each provider by how far its billing behavior sits from its peers and send the strangest ones to a reviewer first. I built this on real Medicare claims. On held-out providers the screen finds fraud at about 2.1 times the base rate using only behavioral features, and each flagged provider comes with a short, generated explanation of why it was flagged. I also trained a supervised model as a baseline, and I am honest about where the unsupervised approach wins and where it does not. The report ends with three options Cotiviti could actually run with.

Keywords: anomaly detection, payment integrity, unsupervised learning, Medicare claims, HDBSCAN, Isolation Forest, explainable AI

1. The Problem

Treatment, Payment, and Operations produce a lot of claims data, and within Payment the hard question is which billing is legitimate and which is not. There are two ways to attack this. A supervised model learns from confirmed-fraud labels and classifies new providers. An unsupervised model ignores labels, learns what normal billing looks like, and flags whatever does not fit. I went with the second approach on purpose. In the real world confirmed labels are rare, they show up long after the fraud happened, and they only describe schemes someone already caught. A model trained on last year's fraud cannot see this year's.

The numbers make the case for why this matters. CMS reported a Medicare fee-for-service improper-payment rate of 7.66 percent, about \$31.7 billion, in fiscal year 2024, and 6.55 percent, about \$28.8 billion, in 2025 [1], [2]. Not all of that is fraud, a lot of it is missing paperwork, but it is the pile that payment-integrity teams have to dig through, and anything that helps them dig the right hole first has value.

2. What Is Changing

A few things are shifting in this space. Teams are moving off fixed rule sets, because once people know the rule they bill just under it. They are leaning more on anomaly methods that can surface new schemes instead of only the ones already in the training data [3]. And they are starting to bolt language models onto the output, not to make the decision, but to turn a pile of numbers into something a reviewer can read and act on. CMS itself has been adding AI to its fraud work and pushing detection earlier, toward checking claims before they are paid rather than chasing money afterward [2].

3. What I Built

The pipeline has four parts: features, detection, an explanation step, and a supervised baseline to compare against.

3.1. Features

I roll every provider's inpatient and outpatient claims up into one row. The important choice here is that I describe each provider with ratios, dollars per claim, claims per patient, codes per claim, claims per physician, instead of raw totals. If you use raw totals the model mostly learns "this is a big hospital," which is

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not fraud, it is just size. I also added two time-series features from the claim dates, how bursty a provider’s billing is and how much it swings month to month, since some schemes bill in short spikes. Missing values get filled with the median, not zero, because a missing length-of-stay and a zero length-of-stay are not the same thing.

3.2. Detection

I use two detectors that look at the data differently. HDBSCAN clusters providers by density and marks the ones in sparse regions as outliers [4]. Isolation Forest gives every provider a continuous score based on how easily it can be split off from the rest [5]. I combine them by averaging their percentile ranks, so neither needs a hand-picked weight. UMAP only shows up for drawing the 2D map [6]; I do not cluster on it.

$$s(x) = \frac{1}{2} [\text{rank}_{\text{IF}}(x) + \text{rank}_{\text{HDB}}(x)] \quad (1)$$

A provider lands in the review queue if its combined score $s(x)$ is in the top 10 percent. That mirrors how a real team works: limited people, so look at the worst cases first.

3.3. Explanations

For any flagged provider, I take the features that deviate most from peers and hand those numbers to an LLM (LLaMA 3.1 on Groq), which writes a few sentences on why it looks off and what a reviewer should check. The model never decides anything. The evidence, how many standard deviations each feature sits from the peer average, is computed by the pipeline; the model just puts it into English. If there is no API key it falls back to a templated version of the same thing, so the demo never breaks.

4. Results

I evaluated this the honest way: fit the detectors on one set of providers, score a held-out set they never saw. The base fraud rate is 9.3 percent. Table 1 shows the screen next to a Random Forest trained on the same features.

Table 1: Out-of-sample results on held-out providers.

Metric	My screen	Supervised RF
Lift over base (top 10%)	2.1×	6.3×
Recall at top 10%	43%	—
AUC	—	0.94
Stability (5 seeds)	2.14× ± 0.30	—

I want to be straight about one number. My first version reported 4.3 times lift, and I was briefly happy about it. Then I realized two things were wrong. I was detecting on raw size features, so the model was mostly flagging big providers, and I was testing on the same data I fit on. Once I fixed both, switched to size-free ratios and a real train/test split, the lift dropped to 2.1. That drop is the honest part. Most of the 4.3 was the model noticing which providers were large, not which were committing fraud.

The supervised model scores much higher here, and that is expected, it has the labels. I am not pretending the unsupervised screen beats it on this dataset. The point is that in production you usually do not have fresh labels, and a supervised model only catches the fraud it was trained on. The unsupervised screen is there to catch the stuff nobody has labeled yet. Run both.

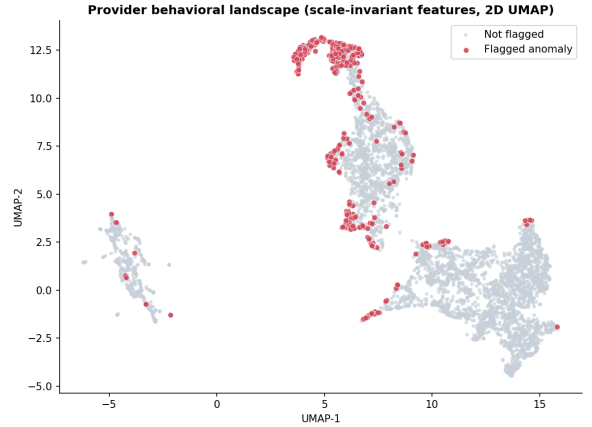


Figure 1: Flagged providers sit in a distinct part of the behavioral map.

5. Where It Helps and Where It Breaks

Where it helps. It is a triage tool. A reviewer only has so many hours, and working the top of this list means hitting fraud roughly twice as often as working the list at random. It can catch new schemes before any label exists, and the written explanation saves the reviewer time and leaves a paper trail for why a provider was looked at.

Where it breaks. An anomaly is not the same as fraud, and I ran into this directly: the single highest-scored provider in my run is not fraud, it is a legitimate high-intensity specialist. Without grouping providers by specialty, those get flagged. A careful fraudster who bills close to average slips right past an outlier detector. Averaging a provider down to one row hides someone who

bills clean 95 percent of the time. And none of this touches the real production requirements: access control, audit logs, drift monitoring, and keeping a human on the actual decision.

6. Three Options for Cotiviti

A. Use it as a triage layer. Put the screen in front of the existing review queue to rank providers by risk. It adds to the current process instead of replacing rules, and it brings in schemes the current models miss.

B. Add the explanation layer. Pair detection with a controlled LLM step that turns the evidence into a reviewer-ready writeup and a record for any action taken. Faster review, cleaner audit trail.

C. Fix the false positives. Group providers by specialty and region and go down to the claim line. This is the work that would push precision up before scaling, and it directly targets the specialist problem above.

What I would do. Start small: A and B together on one claim type, measure how much faster and more accurate reviewers get versus today, then invest in C before going wide.

7. Conclusion

Anomaly detection fits payment integrity well exactly where labels are missing and schemes keep changing. Even a simple version, evaluated honestly, already finds fraud at a useful rate from behavior alone, and an explanation layer makes the output something a reviewer can use. For Cotiviti the near-term win is a screen-and-explain layer on top of existing review. The longer game is moving it earlier, before the payment goes out.

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